

GENE 8160: ADVANCED EVOLUTIONARY GENETICS

Spring 2013, 3 credits

Class meetings: Monday and Wednesday 9:05-11:00 am, Life Sciences Room C114

Prerequisite: GENE 8150 or GENE 6000, or instructor permission

Instructors: Kelly Dyer and John Wares

Contact Information and Office Hours

KD: Email: kdyer@uga.edu, Phone: 706-542-3154, Office: Life Sciences C226

Office hours: By appointment (please email)

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Office hours: By appointment (please email)

Course description and goals

This course explores the process of evolution at the molecular level. We will use basic evolutionary principles to explore how DNA changes through time; to infer population and species history from DNA sequences; to determine what forces have shaped the evolution of genes and genomes; and to understand the relationship between molecular evolution and phenotypic evolution. This field is large and diverse, and combines population genetics, classical genetics, ecology, phylogenetics, molecular biology, and genomics. The goals of this class are to introduce you to some of the main questions, give you a feel for the kinds of data used, and expose you to the tools that are used to answer questions.

Texts and assigned readings

Reading for this course will be primarily articles and reviews from the primary literature.

Articles will be available as PDF files on eLearning Commons.

You will want to keep the following book handy to use as a basic reference:

Principles of Population Genetics, 4th edition (2008) by D. L. Hartl and A. G. Clark.

Other useful resources

Molecular Ecology, J. Freeland. Wiley.

Population Genetics, A concise guide. 2nd ed. J. Gillespie. Johns Hopkins Press

Fundamentals of Molecular Evolution, 2nd ed. By D. Grauer and W.-H. Li. Sinauer Press.

Computational Molecular Evolution, Z. Yang. Oxford University Press.

Class format, Participation, and Attendance

This class will combine lectures, discussions, and computer activities. We expect that you come to class prepared and ready to engage in whatever activities we will be doing that day. Regular attendance in lecture is expected and necessary for learning the material, which is cumulative in nature. After two unexcused absences, each additional absence will result in the loss of 1% from your final average, except in the case of a documented medical or extreme personal emergency.

Grading

Grades are A-F and will be based on three exams (25% each), and assignments/presentations (25%).

Lecture Handouts

A pdf copy of the lecture slides will be available before class on eLearning Commons.

Exams

Three non-cumulative exams will be given on 2/6/13, 3/20/13, and 4/29/13 during class. Makeup exams will only be given in the case of a documented medical or extreme personal emergency. All make-up exams will be oral examinations. If you submit a request for a regrade, your entire exam will be regraded.

Short assignments

Short assignments may come in many flavors, depending on the nature of the material. In addition to traditional problem sets, throughout the semester you will write several 'response' papers (~ 1 page typed and in essay format) to journal articles or class discussions. For journal articles, these should contain (1) the main question is being addressed, (2) what the authors did, (3) what the authors found, (4) the significance of the findings, and (5) a question you had about the paper (e.g. What left you baffled? What techniques were new? Were the conclusions inappropriate? Was the design inadequate?) and (6) two to three questions or topics for discussion. Unless you have been granted an extension ahead of time, a penalty of 10% per day will be applied to assignments turned in after the beginning of class on the day it is due.

Discussions

Frequently we will spend part of a class period discussing a paper from the primary literature. When you present a paper, you will introduce the paper in a 10-15 minute presentation and then lead the discussion of the paper. You will likely want to come prepared with some 'kick off' questions for discussion. You will be evaluated based on if you introduced the reading in a knowledgeable manner, presented data/figures/results clearly and concisely, engaged the class in insightful discussion, effectively answered questions, and in general helped to explain the science. You need to know the paper you are presenting backward and forward and inside out, and you probably will want to read related papers as well.

Electronics

Please turn your phone off during class and office hours.

Religious Holidays

If you plan to miss a class because of a religious holiday, you must talk to one of the instructors at least fourteen days prior to the class to make arrangements to make-up missed work.

Honor Code

All students will abide by the UGA Student Honor Code: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." All suspected incidences of academic dishonesty, as outlined under prohibited conduct in the UGA academic honesty policy (<http://www.uga.edu/ovpi/honesty/sect05.htm>), will be treated following UGA guidelines (http://www.uga.edu/ovpi/honesty/culture_honesty.htm).

Disclaimer

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

GENE 8160: Tentative course schedule

Class	Date		Tentative Topic
			Unit 1: Molecular Ecology
1	M 1/7	JW	Standing diversity
2	W 1/9	JW	Effective population size
3	M 1/14	JW	The Coalescent and Drift
4	W 1/16	JW	The Coalescent continued
	M 1/21		No class: MLK Holiday
5	W 1/23	JW	Model testing
6	M 1/28	JW	Molecular population genetics
7	W 1/30	JW	Population structure
8	M 2/4	JW	Phylogeography
9	W 2/6	JW	
10	M 2/11	JW	EXAM #1
			Unit 2: Phylogenetics
11	W 2/13	KD	Introduction and models of DNA evolution
12	M 2/18	KD	Maximum likelihood
13	W 2/20	KD	Bayesian methods
14	M 2/25	KD	Rates and dates
15	W 2/27	KD	Gene trees vs. species trees
16	M 3/4	KD	Case studies
17	W 3/6	KD	Comparative method
	M 3/11		No class: Spring Break
	W 3/13		No class: Spring Break
18	M 3/18	KD	Selection along a phylogeny
19	W 3/20	KD	Case studies
20	M 3/25	KD	EXAM #2
			Unit 3: Evolutionary genomics
21	W 3/27	KD	Intro to population genomics
22	M 4/1	KD	Candidate locus tests
23	W 4/3	KD	Genome scans
24	M 4/8	KD	Adaptation genomics
25	W 4/10	KD	Case studies
26	M 4/15	KD	Duplications
27	W 4/17	KD	Non-coding DNA evolution
28	M 4/22	KD	Genome evolution
29	W 4/24	KD	Case studies
30	M 4/29	KD	EXAM #3